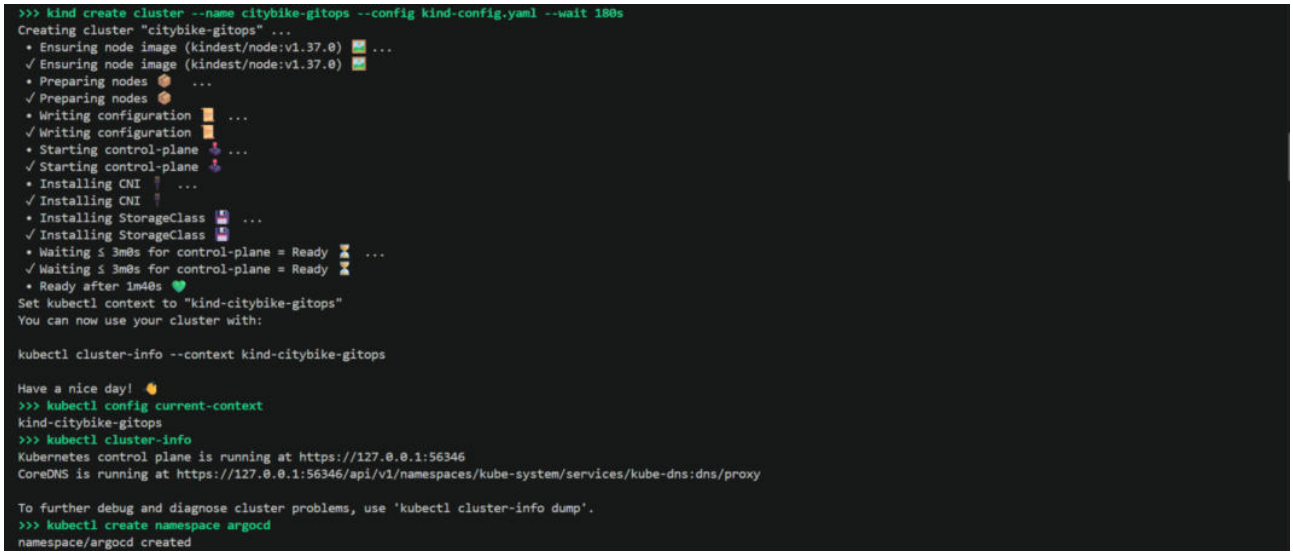
LAB 05 EVIDENCE KELYAN IHINACK

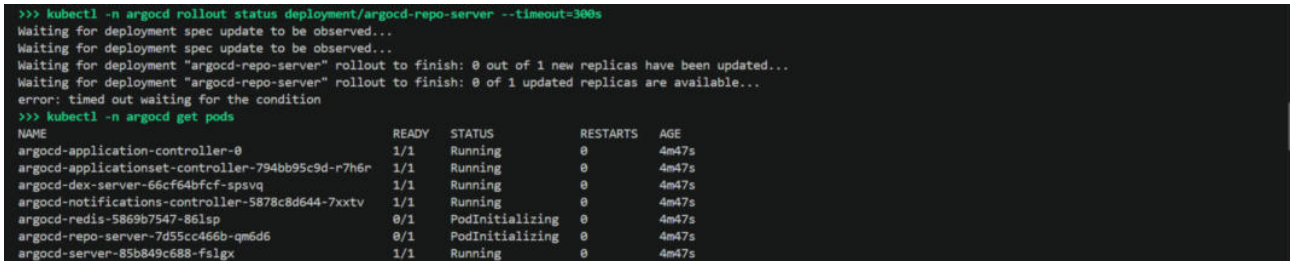
Kubernetes and GitOps with Argo CD. Repo: github.com/kelyanzzz/citybike-k8s-gitops (public). Solo run: I played both Student A and Student B.

Mission 0

Disposable kind cluster created. The first try timed out at "Starting control-plane"; the second try worked. The context is kind-citybike-gitops, then Argo CD is installed.

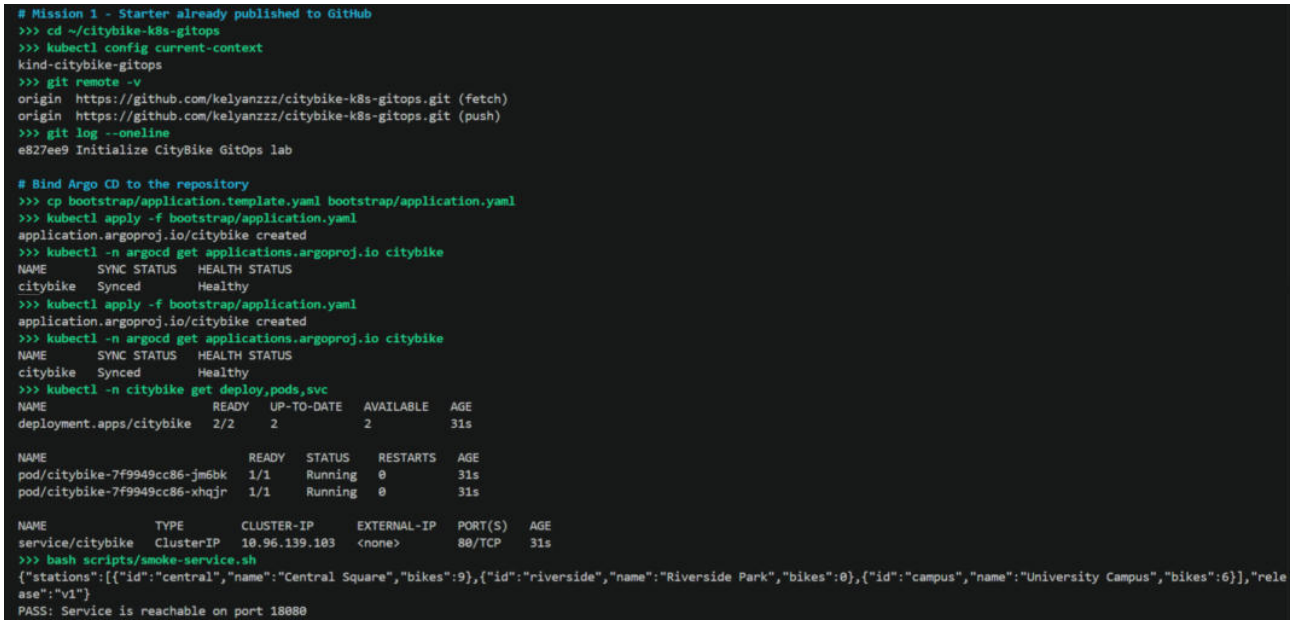


Argo CD pods: the repo-server was still pulling its image when the 300 s check ended. It became Ready a few seconds later.



Mission 1

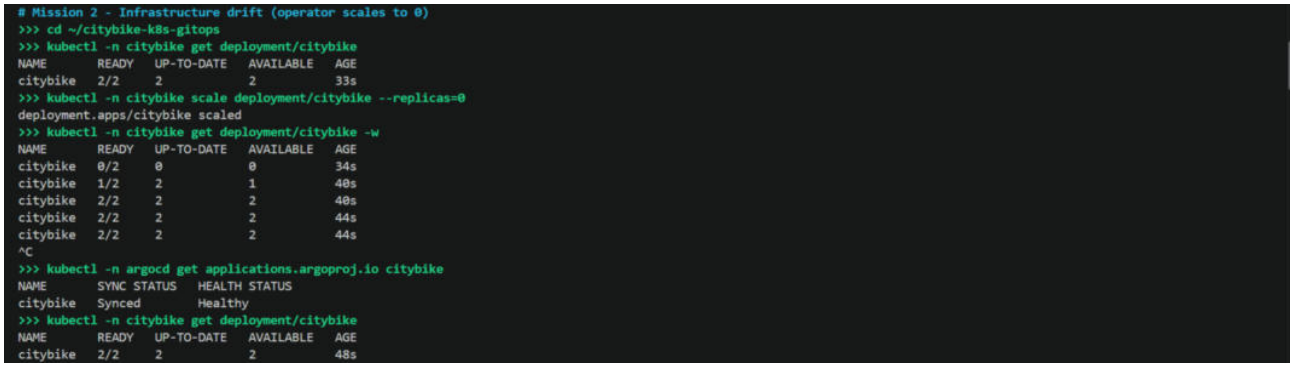
Starter pushed to my public repo. bootstrap/application.yaml points Argo CD to it. The app is Synced and Healthy, 2 pods Ready, and the smoke test on /api/stations passes.



- Kustomize turns app/server.js into a ConfigMap with a hash in its name. A code change gives a new name, so the Deployment changes and Kubernetes rolls out new pods.
- The manifests in Git are the source of truth: Argo CD makes the cluster match them.
- Synced means the live objects match Git. It does not mean users are served correctly (see Mission 3).

Mission 2

Operator scales to 0 by hand. Argo CD self-heal puts it back to 2 replicas in about 6 seconds, without anyone restoring it.



Timeline: 15:52:00 2/2 Ready, 15:52:00 scaled to 0, 15:52:10 back to 2/2 Ready, app Synced/Healthy.

- kubectl scale does not persist because Git still says 2 replicas and selfHeal reverts any live change.
- Without selfHeal, the app would show OutOfSync and stay at 0 until someone syncs.

Mission 3

Bad change on feature/broken-service: targetPort 3000 to 3999, pushed and merged to main.

```
# Mission 3 - A bad Git change breaks access
>>> cd ~/citybike-k8s-gitops
>>> git switch -c feature/broken-service
Switched to a new branch 'feature/broken-service'
>>> git diff
diff --git a/gitops/service.yaml b/gitops/service.yaml
index 8f56780..e2e290d 100644
--- a/gitops/service.yaml
+++ b/gitops/service.yaml
@@ -11,5 +11,5 @@ spec:
   ports:
     - name: http
       port: 80
-      targetPort: 3000
+      targetPort: 3999
       protocol: TCP
>>> git commit -am "Change Service targetPort to 3999"
[feature/broken-service 3522a06] Change Service targetPort to 3999
 1 file changed, 1 insertion(+), 1 deletion(-)
>>> git push -u origin feature/broken-service
remote:
remote: Create a pull request for 'feature/broken-service' on GitHub by visiting:
remote:      https://github.com/kelyanzzz/citybike-k8s-gitops/pull/new/feature/broken-service
remote:
branch 'feature/broken-service' set up to track 'origin/feature/broken-service'.
To https://github.com/kelyanzzz/citybike-k8s-gitops.git
 * [new branch]      feature/broken-service -> feature/broken-service

# Reviewed, merged to main
>>> git switch main && git merge feature/broken-service && git push
Your branch is up to date with 'origin/main'.
Switched to branch 'main'
Updating e827ee9..3522a06
Fast-forward
 gitops/service.yaml | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
To https://github.com/kelyanzzz/citybike-k8s-gitops.git
   e827ee9..3522a06  main -> main
>>> kubectl -n argocd annotate applications.argoproj.io citybike argocd.argoproj.io/refresh=hard --overwrite
application.argoproj.io/citybike annotated
>>> kubectl -n argocd get applications.argoproj.io citybike
NAME       SYNC STATUS   HEALTH STATUS
citybike   Synced        Healthy
```

Argo CD is Synced/Healthy and the pods are Ready, but the smoke test fails. The Service sends traffic to port 3999, while the app listens on 3000.

```
>>> kubectl -n argocd get applications.argoproj.io citybike
NAME       SYNC STATUS   HEALTH STATUS
citybike   Synced        Healthy
>>> kubectl -n citybike get pods,svc,endpointslice
NAME                                 READY   STATUS    RESTARTS   AGE
pod/citybike-7f9949cc86-4bf7r   1/1     Running   0          27s
pod/citybike-7f9949cc86-x7mwb   1/1     Running   0          27s
citybike   Synced        Healthy
>>> kubectl -n citybike get pods,svc,endpointslice
NAME                                 READY   STATUS    RESTARTS   AGE
pod/citybike-7f9949cc86-4bf7r   1/1     Running   0          27s
pod/citybike-7f9949cc86-x7mwb   1/1     Running   0          27s

NAME              TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
service/citybike   ClusterIP   10.96.139.103   <none>        80/TCP    61s

NAME                                          ADDRESSTYPE   PORTS   ENDPOINTS                 AGE
endpointslice.discovery.k8s.io/citybike-hz8bs   IPv4          3999    10.244.0.14,10.244.0.15   61s
>>> kubectl -n citybike describe svc citybike
Name:                     citybike
Namespace:                citybike
Labels:                   app.kubernetes.io/name=citybike
Annotations:              argocd.argoproj.io/tracking-id: citybike:/Service:citybike/citybike
Selector:                 app.kubernetes.io/name=citybike
Type:                     ClusterIP
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.96.139.103
IPs:                      10.96.139.103
Port:                     http  80/TCP
TargetPort:               3999/TCP
Endpoints:                10.244.0.14:3999,10.244.0.15:3999
Session Affinity:         None
Internal Traffic Policy:  Cluster
Events:                   <none>
>>> bash scripts/smoke-service.sh
FAIL: Service is unreachable. Port-forward details:
Forwarding from 127.0.0.1:18080 -> 3999
Forwarding from [::1]:18080 -> 3999
Handling connection for 18080
E1106 15:52:29.429717   34660 portforward.go:424] "Unhandled Error" err="an error occurred forwarding 18080 -> 3999: error forwarding port 3999 to pod 69b2d99e8b04cabb534bab9ff078f666f66ae342d80e1323007c3ae0af6cbb24, uid : failed to execute portforward in network namespace \"/var/run/netns/cni-88e05ce2-62f3-ecf8-59c9-f37b57ff6ebd\": failed to connect to localhost:3999 inside namespace \"69b2d99e8b04cabb534bab9ff078f666f66ae342d80e1323007c3ae0af6cbb24\", IPv4: dial tcp4 127.0.0.1:3999: connect: connection refused IPv6 dial tcp6 [::1]:3999: connect: connection refused "
error: lost connection to pod
```

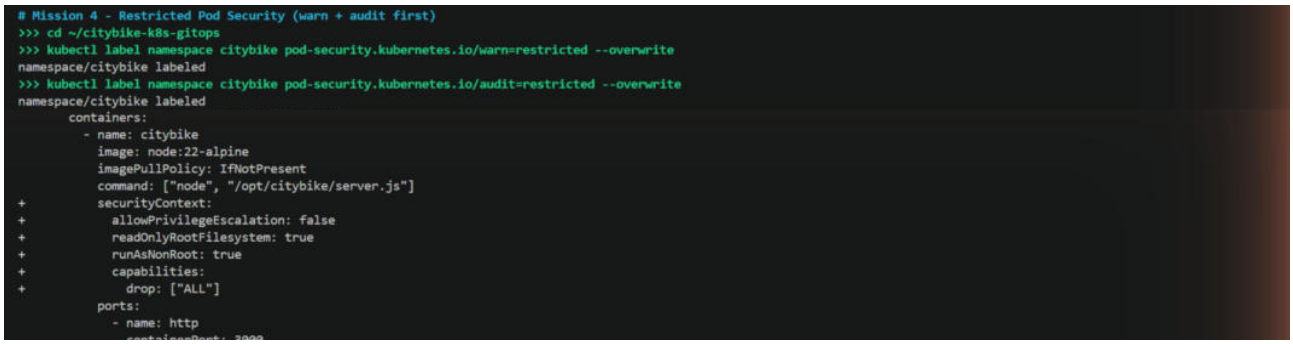
Fix: git revert, new commit on main, Argo reconciles. TargetPort is back to 3000 and the smoke test passes.



- Pods stay Ready because the probes call the container directly on port 3000, not through the Service.
- Synced does not guarantee clients are served: Argo applied exactly what Git said, and Git was wrong.
- Root cause: Service targetPort 3999 does not match containerPort 3000 (EndpointSlice shows port 3999).

Mission 4

Namespace in warn and audit mode first. The hardened Deployment goes through a branch and a merge.



Rollout succeeds, then enforce=restricted is turned on. The warning only lists the old pods that were terminating.

```
>>> git commit -am "Harden Deployment for Restricted Pod Security"
[feature/restricted-security 770cb4c] Harden Deployment for Restricted Pod Security
 1 file changed, 13 insertions(+)
>>> git push -u origin feature/restricted-security
remote:
remote: Create a pull request for 'feature/restricted-security' on GitHub by visiting:
remote:      https://github.com/kelyanzzz/citybike-k8s-gitops/pull/new/feature/restricted-security
remote:
branch 'feature/restricted-security' set up to track 'origin/feature/restricted-security'.
To https://github.com/kelyanzzz/citybike-k8s-gitops.git
 * [new branch]      feature/restricted-security -> feature/restricted-security

# Reviewed, merged to main
>>> git switch main && git merge feature/restricted-security && git push
Your branch is up to date with 'origin/main'.
Switched to branch 'main'
Updating bdbdc67..770cb4c
Fast-forward
 gitops/deployment.yaml | 13 ++++++++++++
 1 file changed, 13 insertions(+)
To https://github.com/kelyanzzz/citybike-k8s-gitops.git
   bdbdc67..770cb4c  main -> main
>>> kubectl -n argocd annotate applications.argoproj.io citybike argocd.argoproj.io/refresh=hard --overwrite
application.argoproj.io/citybike annotated
>>> kubectl -n citybike rollout status deployment/citybike --timeout=180s
Waiting for deployment "citybike" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "citybike" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "citybike" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "citybike" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "citybike" rollout to finish: 1 old replicas are pending termination...
deployment "citybike" successfully rolled out

# Only after a successful rollout: enforce
>>> kubectl label namespace citybike pod-security.kubernetes.io/enforce=restricted --overwrite
Warning: existing pods in namespace "citybike" violate the new PodSecurity enforce level "restricted:latest"
Waiting for deployment "citybike" rollout to finish: 1 old replicas are pending termination...
deployment "citybike" successfully rolled out

# Only after a successful rollout: enforce
>>> kubectl label namespace citybike pod-security.kubernetes.io/enforce=restricted --overwrite
Warning: existing pods in namespace "citybike" violate the new PodSecurity enforce level "restricted:latest"
Warning: citybike-7f9949cc86-4bf7r (and 1 other pod): allowPrivilegeEscalation != false, unrestricted capabilities, runAsNonRoot != true, seccompProfile
namespace/citybike labeled
```

Release v2 with APP_RELEASE=v2: the new pods are admitted under enforce=restricted, 2 Ready, smoke passes with release v2.

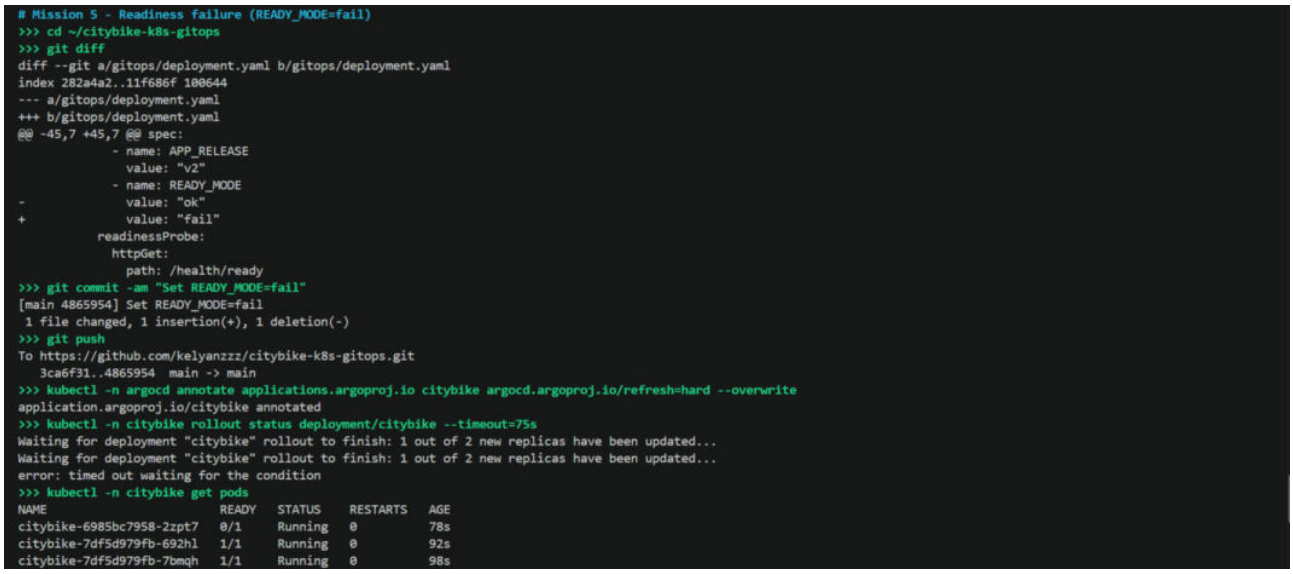
```
# Mission 4 - New release admitted under enforce=restricted
>>> cd ~/citybike-k8s-gitops
>>> git diff
diff --git a/gitops/deployment.yaml b/gitops/deployment.yaml
index e420263..282a4a2 100644
--- a/gitops/deployment.yaml
+++ b/gitops/deployment.yaml
@@ -43,7 +43,7 @@ spec:
             containerPort: 3000
           env:
             - name: APP_RELEASE
-              value: "v1"
+              value: "v2"
             - name: READY_MODE
               value: "ok"
           readinessProbe:
>>> git commit -am "Release v2 (APP_RELEASE=v2)"
[main 3ca6f31] Release v2 (APP_RELEASE=v2)
 1 file changed, 1 insertion(+), 1 deletion(-)
>>> git push
To https://github.com/kelyanzzz/citybike-k8s-gitops.git
   770cb4c..3ca6f31  main -> main
>>> kubectl -n argocd annotate applications.argoproj.io citybike argocd.argoproj.io/refresh=hard --overwrite
application.argoproj.io/citybike annotated
>>> kubectl -n citybike rollout status deployment/citybike --timeout=180s
Waiting for deployment "citybike" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "citybike" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "citybike" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "citybike" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "citybike" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "citybike" rollout to finish: 1 old replicas are pending termination...
deployment "citybike" successfully rolled out
>>> kubectl -n citybike get pods
NAME                        READY   STATUS        RESTARTS   AGE
citybike-77d9f86db5-6nt2f   1/1     Terminating   0          25s
citybike-77d9f86db5-dp6fq   1/1     Terminating   0          31s
citybike-7df5d979fb-692hl   1/1     Running       0          7s
citybike-7df5d979fb-7bmqh   1/1     Running       0          13s
>>> bash scripts/smoke-service.sh
NAME                        READY   STATUS        RESTARTS   AGE
citybike-77d9f86db5-6nt2f   1/1     Terminating   0          25s
citybike-77d9f86db5-dp6fq   1/1     Terminating   0          31s
citybike-7df5d979fb-692hl   1/1     Running       0          7s
citybike-7df5d979fb-7bmqh   1/1     Running       0          13s
>>> bash scripts/smoke-service.sh
{"stations":[{"id":"central","name":"Central Square","bikes":9},{"id":"riverside","name":"Riverside Park","bikes":0},{"id":"campus","name":"University Campus","bikes":6}],"rele
ase":"v2"}
PASS: Service is reachable on port 18080
```

- runAsNonRoot and runAsUser 1000: no root inside the container.
- seccomp RuntimeDefault: blocks dangerous system calls.
- allowPrivilegeEscalation false and drop ALL capabilities: no gaining more rights.

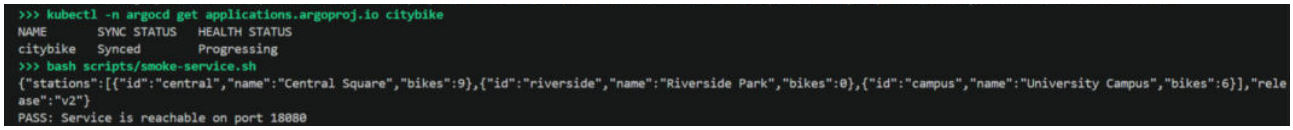
- readOnlyRootFilesystem: an attacker cannot change files. No service account token mounted.

Mission 5

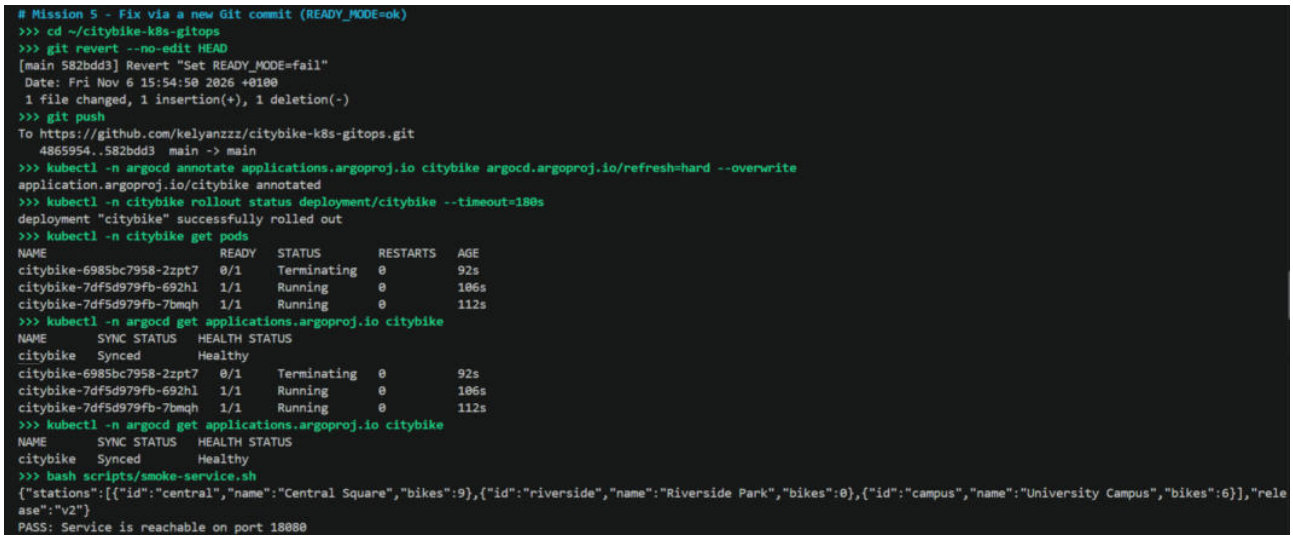
READY_MODE=fail pushed to main. The new pod stays 0/1 (readiness returns 503) and the rollout times out after 75 s.



Argo shows Progressing. Because maxUnavailable=0, the old pods keep serving and the smoke test still passes.



Fix with a new commit (revert to READY_MODE=ok): rollout succeeds and the app is Synced/Healthy again.



- /health/live says the process is alive; /health/ready says it can take traffic.
- Keeping them separate means a not-ready pod gets no traffic but is not killed and restarted for nothing.

Mission 6

Final state: context kind-citybike-gitops, Synced/Healthy, 2/2 Ready, endpoints on port 3000. Every change is a Git commit.



```
# Mission 6 - Evidence
>>> cd ~/citybike-k8s-gitops
>>> bash scripts/status.sh

--- Cluster context ---
kind-citybike-gitops

--- Argo Application ---
NAME       SYNC STATUS   HEALTH STATUS
citybike   Synced        Healthy

--- Workloads ---
NAME                     READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/citybike   2/2     2            2           3m37s

NAME                            READY   STATUS        RESTARTS   AGE
pod/citybike-6985bc7958-2zpt7   0/1     Terminating   0          94s
pod/citybike-7df5d979fb-692hl   1/1     Running       0          108s
pod/citybike-7df5d979fb-7bmqh   1/1     Running       0          114s

NAME              TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
service/citybike   ClusterIP   10.96.139.103   <none>        80/TCP    3m38s

--- Endpoints ---
NAME             ADDRESSTYPE   PORTS   ENDPOINTS                                 AGE
citybike-hz8bs   IPv4          3000    10.244.0.18,10.244.0.19,10.244.0.20   3m38s
>>> git log --oneline --graph --all
* 582bdd3 Revert "Set READY_MODE=fail"
* 4865954 Set READY_MODE=fail
* 3ca6f31 Release v2 (APP_RELEASE=v2)
* 770cb4c Harden Deployment for Restricted Pod Security
* bdbdc67 Revert "Change Service targetPort to 3999"
* 3522a06 Change Service targetPort to 3999
* e827ee9 Initialize CityBike GitOps lab
```

- kubectl apply in production is not acceptable here: Argo would revert it, and it leaves no review or history.
- An app can be Synced and still fail: Synced only compares live state with Git, not whether users are served.
- No secrets were committed. bootstrap/application.yaml is ignored by Git.